Hypertrophic Osteoarthropathy Presenting as Unilateral Cellulitis with Successful Treatment Using Pamidronate Disodium

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ABSTRACT

Hypertrophic pulmonary osteoarthropathy is a paraneoplastic syndrome seen in patients with lung cancer. This condition is characterized by the presence of digital clubbing, periosteal thickening, synovial thickening, and severe pain of the affected joints. Other syndromes exhibiting clubbing may or may not have underlying diseases causing their manifestation. An example is primary hypertrophic osteoarthropathy, or pachydermoperiostosis. While clubbing makes up part of the clinical picture in both hypertrophic pulmonary osteoarthropathy and hypertrophic osteoarthropathy, the latter has no underlying disease associations. Rather, primary hypertrophic osteoarthropathy is familial, idiopathic, and has a chronic course often beginning during puberty in males. Secondary hypertrophic osteoarthropathy is an acquired form of clubbing that is classically associated with lung disease. However, it has also been associated with diseases of the heart, liver, and intestines. In the setting of pulmonary malignancy, secondary hypertrophic osteoarthropathy is known as hypertrophic pulmonary osteoarthropathy. Hypertrophic pulmonary osteoarthropathy has a distinct constellation of clinical findings that includes intractable pain often refractory to treatments other than resolution of the underlying disease process. The authors herein report a case of hypertrophic pulmonary osteoarthropathy masquerading as recurrent lower extremity cellulitis with chronic hand and foot pain in the setting of pulmonary malignancy that responded dramatically to intravenous pamidronate disodium (a bisphosphonate). Given the rarity of hypertrophic osteoarthropathy associated with lung cancer and the difficulty with pain management in such circumstances, the authors present the following case in which pain was mitigated by treatment with bisphosphonate therapy.
(*J Clin Aesthet Dermatol.* 2012;5(9):37–46.)

Hypertrophic pulmonary osteoarthropathy (HPOA) is a paraneoplastic syndrome seen in the setting of pulmonary malignancy that is characterized by digital clubbing, hyperproliferation of skin, periosteal proliferation and inflammation of the long bones, and synovitis.[1] In cases of hypertrophic osteoarthropathy (HOA) secondary to underlying disease, treatment is aimed at either resolution of the primary cause or symptomatic relief. To date, only a handful of other publications have presented cases in which bisphosphonate therapy provided effective symptomatic relief in patients with paraneoplastic-associated secondary HOA.[2] The incidence of HPOA is relatively rare, with a frequency of about 5 to 10 percent among patients with intrathoracic malignancies.[3] However, some studies have reported that up to 90 percent of adults presenting with HOA have, or will develop, a malignancy, pulmonary or otherwise.[4] Thus, prompt and accurate diagnosis of HOA in patients without a history of malignancy is of paramount importance to exclude disease. Successful symptomatic management with pamidronate disodium in a patient with HPOA presenting as cellulitis has only been reported twice elsewhere in the English literature.[5,6] For education and review, the authors present a case of HPOA characterized by periosteal bone formation, arthritis, and clubbing of the digits that masqueraded as lower extremity cellulitis in a patient with metastatic lung cancer.

DISCLOSURE: The authors report no relevant conflicts of interest.
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Figure 1. Erythematous plaque on the right shin without surrounding edema or associated warmth



Figure 2. Clubbing of the digits on the right hand





Figures 3A and 3B. A) Periungual erythema of the digits on the right hand at initial presentation; B) periungual erythema of the toes on the right foot at initial presentation

CASE REPORT

A 63-year-old Caucasian man with a past medical history of stage IV non-small cell lung cancer (NSCLC) and a 40 pack-year smoking history presented to the emergency department with complaints of a recurrent burning and painful right lower leg for several months. The patient reported multiple exacerbations of redness associated with swelling on his right shin in the preceding months. In the days prior to this presentation, the patient completed his last cycle of chemotherapy with palonosetron and docetaxel and was currently on palliative care for intractable metastatic disease. On review of systems, the patient noted generalized joint pain that was most severe in the distal hands and feet as well as right upper quadrant abdominal pain with non-bloody diarrhea unrelieved by oral over-the-counter symptomatic care.

On physical examination, the patient had a fixed, well-demarcated patch of erythema with mild edema on his right shin extending from the ankle to the knee that was tender to palpation and warm (Figure 1). Digital clubbing was noted along with periungual erythema of the hands and feet (Figures 2, 3a, and 3b). Range of motion of the



Figure 4. Marked decrease in erythema of the right shin 48 hours after 90mg intravenous pamidronate infusion



Figure 5. Resolution of erythema on the right hand 48 hours after 90mg intravenous pamidroante infusion

upper and lower extremities was limited due to severe joint pain bilaterally. Following a comprehensive physical examination, more detailed questioning revealed multiple previous hospitalizations for presumed cellulitis with only mild relief from intravenous antibiotic therapy and pain management with narcotics, such as oxycodone and hydromorphone. Multiple evaluations for distal vein thrombosis were negative and compression stocking therapy with topical corticosteroid and emollients for stasis dermatitis provided no relief. The patient described his right shin as having recurrent "flares" of redness and pain that was fixed to the same location. He also described a sensation of "lumpiness" in his hands and feet, likening the pain in his feet to "walking on marbles." Prior to this hospitalization, his outpatient medications included the following: oxycodone 325/5mg four times daily, prednisone 10mg daily, celecoxib 200mg twice daily, and folic acid 1mg daily. His family and social histories were noncontributory. He denied any alcohol or illicit drug use.

Given the findings of digital clubbing accompanied by diffuse joint pain and skin erythema of the hands and feet in a patient with biopsy-proven NSCLC, the clinical diagnosis of HPOA was made. The patient was admitted to the hospital for supportive care and a single 90mg dose of intravenous pamidronate was recommended. Within 48 hours of therapy, the patient and family noted a marked improvement in the right shin erythema, finger erythema, and polyarthropathy (Figures 4 and 5). Within 72 hours, the symptoms had completely resolved. Radiation oncology was consulted to address the new hepatic metastasis thought to be responsible for the patient's abdominal pain and diarrhea. He was deemed a candidate for external-beam radiation therapy following stabilization and pain management. The patient was subsequently discharged for outpatient palliative care treatment following a 10-day hospitalization.

DISCUSSION

Digital clubbing is a clinical finding characterized by bulbous enlargement of the distal phalanges and may be congenital, idiopathic, or acquired.[7] In patients with hypertrophic osteoarthropathy (HOA), clubbing is accompanied by hyperproliferation of skin, periosteal proliferation and inflammation of the long bones, and synovitis.[1,3] Primary HOA, also known as pachydermoperiostitis, is a genetically inherited disorder that affects male patients nine times more frequently than female patients. Commonly presenting during puberty, primary HOA is characterized by clubbing, periosteal proliferation, and synovitis.[7] Pachydermia (thickening of the facial skin and/or scalp), eyelid ptosis, polyarthritis, cutis verticis gyrata (thickening and corrugation of the skin overlying the scalp), seborrhea, and hyperhidrosis are other clinical findings associated with primary HOA.[8,9] Interestingly, disease progression of primary HOA stops at the end of adolescence.[3] It is important to note, however, that while primary HOA is not associated with underlying disease at presentation, a latency of up to 20 years has been reported in the onset of systemic diseases that are known to cause secondary HOA.[10]

Secondary HOA is an acquired syndrome characterized by digital clubbing, arthralgias, and periosteal proliferation of the long bones.[1] Other recognized dermatological features include skin thickening of the face and scalp and hyperhidrosis of the palms and soles.[1,11] The majority of

secondary HOA cases are associated with an intrathoracic malignancy, such as pulmonary carcinoma or lymphoma, characterizing the symptom constellation of secondary HOA as a paraneoplastic syndrome. While many cases of secondary HOA are associated with lung malignancy, the triad of digital clubbing, arthralgias, and periosteal proliferation has also been described in the context of other intrathoracic disease processes, such as infections and cyanotic heart disease. Furthermore, the differential diagnosis of digital clubbing as a sole physical finding includes disorders of the respiratory, cardiovascular, and gastrointestinal systems as well as traumatic and genetic causes. Thus, a comprehensive workup is required to rule out underlying conditions.[12] In the case of clubbing with periosteal proliferation, however, intrathoracic malignancies account for approximately 90 percent of cases.[11]

Etiology. The exact mechanisms underlying the pathogenesis of HOA remain to be elucidated. Primary HOA follows a chronic course and is known to occur without an identifiable primary cause. Secondary HOA, on the other hand, is frequently rapid in its progression and manifests as the result of a specific disease entity.[13–15] Furthermore, secondary HOA presents with tenderness and pain in peripheral long bones and periosteal proliferation. Increased thickening of soft tissues in the distal parts of the arms and legs as well as neurovascular changes in the hands and feet are frequently seen and can be accompanied with redness of the skin.[15] The hypertrophic changes of the bones, joints, and nails that occur in all cases of secondary HOA suggest that a common mechanism is at the basis of disease pathophysiology.

Early theories regarding the pathogenesis of HOA postulated that tissue hypoxia, neural reflexes, genetic influences, and circulating factors, such as growth hormone were influential in disease manifestation.[16–18] However, a growing body of literature now suggests that abnormal tumor production and/or function of vascular endothelial growth factor (VEGF) and platelet-derived growth factor (PDGF) may be central mediators. Normally inactivated by lung tissue, these factors are thought to act selectively on the digital microvasculature causing vasodilatation and increased cellular proliferation. Studies have demonstrated circulating levels of VEGF and PDGF that originate from megakaryocyte and platelet emboli in patients with both primary and secondary HOA.[19–25] In a case report of HPOA occurring in the setting of lung cancer, a 61-year-old man with progressive swelling in both hands, bilateral clubbing of the fingers and toes, and high preoperative plasma levels of VEGF and interleukin (IL)-6 underwent an upper left lobectomy. Rapid postoperative improvement of HOA symptoms and normalization of VEGF and IL-6 was reported within one month of surgical resection. The authors concluded that markedly reduced levels of VEGF and IL-6 following tumor resection correlated with the patient's clinical improvement.[22]

VEGF is a platelet-derived cytokine regulated by tissue hypoxia that acts as a potent angiogenic stimulus, permeability-enhancing factor, and regulator of new bone formation.[26–28] PDGF is a mitogenic protein that increases vascular permeability, promotes monocyte and neutrophil chemotaxis, and stimulates growth and proliferation of vascular smooth muscle cells and fibroblasts.[29] It has been suggested that in lung carcinoma, clubbing and HOA may be influenced by an alteration in the ability of both VEGF and PDGF to interact with platelets and the endothelium.[29–31] Perhaps the most promising hypothesis was put forth by Dickson and Martin[32] who posited that the acropachy (subperiosteal new bone formation) occurring in the nail bed was a result of increased VEGF and PDGF activity arising as a function of platelet aggregation and clumping in this tissue. Their proposal was predicated on evidence showing that megakaryocytes are normally fragmented into platelets in the lung. However, in conditions that disrupt normal pulmonary circulation (i.e., chronic lung inflammation, bronchial tumors, cardiac shunts), whole megakaryocytes would be allowed to enter the systemic circulation and become lodged in the fingertip circulation because of their large size relative to the vasculature in this area. Disruption of the normal acral microvascular circulation would cause shearing of the megakaryocytes creating fragments which would activate the release of growth factors, such as PDGF that can cause a proliferation of collagen and vasculature in the subungual tissue bed.[15,32,33] This hypothesis has become known as the megakaryocyte/platelet theory and has been supported by several other subsequent studies in the literature.[34,35]

In a histological study of five patients with digital clubbing associated with underlying pulmonary disease (4 with bronchogenic carcinoma and 1 with fibrosing alveolitis), an increased number of platelet clumps in the nail-bed capillaries of clubbed digits was found when compared with controls.[35] The authors obtained necropsy nail-bed samples from the five patients postmortem and compared them to 25 samples taken from random controls. Samples were stained with a panel of monoclonal and polyclonal antibodies directed against markers specific to both platelet microthrombi (CD31, CD61, factor VIII) and whole blood microthrombi (glycophorin C). In all five patients with digital clubbing and lung disease, numerous platelet microthrombi were found. In contrast, only 4 of the 25 controls demonstrated "very occasional platelet and whole blood microthrombi." The authors concluded that the presence of platelet microthrombi in the five cases of clubbing associated with underlying pulmonary disease supports the hypothesis that PDGF is important in the pathogenesis of digital clubbing. Thus, it is likely that angiogenesis, subperiosteal new bone formation, and synthesis of the connective-tissue matrix are initiated and propagated by factors within platelet granules that are released upon impaction in the nail bed vasculature during interaction with local endothelium.[31]

A more recent study evaluated the hypothesis that platelet clusters impacting in the distal vasculature mediates clubbing. The researchers used immunohistochemistry to evaluate eight different parameters in the stroma of clubbed digits and compared them to control samples. In comparison

to controls, the stroma of clubbed digits demonstrated statistically significant increases in 5 of the 8 parameters: VEGF (p=0.01), PDGF (p=0.017), microvessel density (p=0.03), hypoxia-inducible factor (HIF)-1 alpha (p=0.004), and HIF-2 alpha (p=0.004).[33] HIF-1-alpha and HIF-2-alpha are upregulated in the setting of hypoxia and serve as transcription factors in the production of growth factors, such as VEGF. This suggests that platelet impaction and aggregation within distal capillaries may result in vascular occlusion and create a hypoxic microenvironment in the stroma that stimulates the production of these HIF factors that consequently causes an upregulation of VEGF and PDGF. This supports the hypothesis that VEGF and PDGF may work synergistically in stimulating the tissue changes that lead to the digital clubbing seen in HOA.

Finally, a study investigating the role of platelets in the pathogenesis of HOA measured circulating levels of endothelin-1 (ET-1), an endothelial cell-derived vasoconstrictive peptide, as well as beta-thromboglobulin (beta-TG) and PDGF, two peptides that are synthesized in the alpha-granules of platelets. The researchers determined the concentration of these peptides in a panel of 21 patients with HOA (6 with primary HOA and 15 with secondary HOA) and found that circulating levels of ET-1, beta-TG and PDGF (p=0.0001, p=0.0001, and p=0.0006, respectively) were significantly higher in both primary and secondary HOA patients when compared with healthy controls.[19] These findings suggest that the pathogenesis and clinical changes observed in patients with HOA may in fact be due to an abnormal interplay between platelets and endolethial cells.

Clinical presentation. As previously described, HOA is a clinical diagnosis that manifests as a triad of periostosis of long bones, arthralgias, and bilateral clubbing of the digits. It may occur as primary or secondary HOA, two conditions with different clinical presentations that affect distinct patient populations. Primary HOA is a genetically inherited disorder known as pachydermoperiostosis. It is characterized by coarse facial features, furrowing and oiliness of the skin of the face, clubbing of the fingers, marked furrowing of the scalp (cutis verticis gyrata), and radiographic periostitis of the distal long bones. In contrast, secondary HOA may result from a number of underlying disease processes including lung cancer, cardiac diseases, and infections.[6,36] The strongest association is seen in pulmonary carcinomas, specifically NSCLC.[37] As a result, secondary HOA typically affects patients within the age cohort most commonly affected by NSCLC—between the ages of 50 and 75 years. The development of secondary HOA may either precede or follow the development of an underlying neoplasm. As such, it may be a harbinger of malignant disease in some patients.[38]

The clinical presentation of primary and secondary HOA varies according to the onset, duration, and evolution of specific signs and symptoms. In primary HOA, disease onset is typically insidious and often asymptomatic. Patients may become accustomed to associated features, such as palmoplantar hyperhidrosis, clubbing of the hands and feet, cutis verticis gyrata, and furrowing and thickening of the facial and forehead skin. As a result, patients may not actively seek medical attention and assume these features to be normal parts of their body image for several years.[39,40] Bone or joint pain with recurrent swelling due to periosteal new bone formation may be the only symptomatic manifestation that initiates a medical workup.



Figure 6. An example of digital clubbing with loss of angle between the nail bed and cuticle seen on lateral view

In secondary HOA, the presence of neoplastic or inflammatory pulmonary conditions are typically associated with rapidly evolving joint pain and swelling of acute onset. Constitutional symptoms, such as fatigue, weight loss, and fever, as well as respiratory symptoms, including cough, hemoptysis, chest pain, and dyspnea, are common. Since the diagnostic criteria for HOA includes clubbing, there will be a loss of the normal 165-degree angle between the cuticle and nail bed that can be seen on lateral views (Figure 6). If apposing nails are placed together, the normal, oblong, diamond-shaped aperture between the two juxtaposed nail beds is lost. Periostosis may be asymptomatic or cause swelling accompanied by severe burning or deep-seated pain that is aggravated with dependency and relieved with elevation of the affected limb.[41,42] Increased connective tissue deposition coupled with vascular proliferation and new periosteal bone formation cause deep-seated pain in the distal extremities.[43] The arthritis associated with secondary HOA is symmetrical, bilateral, and can involve large joints, such as the knee, as well as the small joints of the hands, wrists, elbows, and ankles. Arthritis or dysesthesia of the fingers can present with clumsiness, stiffness, and hyperhidrosis. Range of motion may be affected as well, especially if joint effusions are present. Induration, erythema, dyspigmentation, warmth, and tenderness all represent cutaneous manifestations of underlying swelling, bone formation, and angiogenesis.

Since the periostosis that is characteristic of HOA is hyperemic, bone scanning using ^{99m}Tc diphosphonate is a sensitive, albeit less specific, modality than radiography in assessing osteoblastic changes throughout the course of the disease. Even the most subtle changes in disease progression are often detectable with scintigraphy, allowing pathology to be visualized earlier than would be seen using

plain x-rays. Radiographic evidence of HOA typically involves the appendicular skeleton, often including bilateral periostitis of the joints between the distal interphalanges as well as both the proximal and distal diaphyses in the long bones of the arms and legs.[44]

Differential diagnosis. Generating a differential diagnosis for HOA first requires determining whether it is a primary or secondary process. As described previously, primary HOA or pachydermoperiostosis is a genetically inherited phenomenon mostly affecting males in childhood.[10,45] Although it shares many of the same clinical manifestations as secondary HOA, the onset of disease and presence of associated systemic symptoms make the two relatively easy to differentiate. Again, pachydermoperiostosis is characterized by digital clubbing, periostosis, and dermatological manifestations, such as cutis verticis gyrata, course facial features, furrowing of the eyebrows, ptosis of the eyelid, seborrhea, hyperhidrosis, and skin dyspigmentation.[6,41] In contrast to patients with secondary HOA, cutaneous changes tend to be more prominent than rheumatological complaints in patients with primary HOA. It is important to note that while coarse facial features are more commonly associated with primary HOA, it is by no means disease specific and may be seen in adults with secondary HOA as well. Similar coarse facial features can be seen in acromegaly and should be investigated with appropriate laboratory tests, such as insulin-like growth factor 1 (IGF-1) and growth hormone, along with imaging modalities, such as magnetic resonance imaging (MRI), to rule out pituitary adenomas.

Bilateral and symmetrical joint pain especially of the metacarpophalangeal, proximal interphalangeal, and wrist joints can resemble rheumatoid arthritis (RA). However, in contrast with RA, the joint effusions associated with HOA are relatively noninflammatory, and swelling is typically due to the underlying bone formation and angiogenesis rather than autoimmune-mediated inflammation. Although erythrocyte sedimentation rate (ESR) is usually elevated in both disease processes, rheumatoid factor is not elevated in HOA.[41]

Some cases of HOA may present with a constellation of symptoms similar to polymyalgia rheumatica, a clinical syndrome characterized by severe aching and stiffness of the proximal muscle groups of the neck, shoulders, and pelvic girdle. Patients with polymyalgia rheumatica, however, typically complain of symptoms that are acute, symmetrical, and worse in the morning without associated clubbing or periostosis. Laboratory changes and physical examination with a focused assessment of proximal muscle strength will aid in making the appropriate diagnosis.[46]

Given the significant differences in their respective treatment protocols, it is important to distinguish the osteoarticular manifestations of HOA from skeletal metastases due to a primary malignancy. A review of 4,000 cases of lung cancer in Edinburgh found that 49 patients (1.2%) presented with symptoms of HOA while 63 (1.6%) presented with bone pain that was attributed to skeletal metastases.[47] In contrast with HOA, metastatic disease is less likely to present with symmetrical involvement or be associated with edema.[48]

Finally, thyroid acropachy, an extreme manifestation of autoimmune thyroid disease, is another important condition to consider. Acropachy is characterized by subperiosteal bone formation and noninflammatory swelling of the fingers and dorsum of the hands causing digital clubbing. It can be associated with myxedema resulting in pretibial disease and exophthalmos. Symptoms abate with adequate treatment of the underlying thyroid dysfunction.[41]

The patient described in this case was initially misdiagnosed with cellulitis and stasis dermatitis due to frequent bouts of erythema, redness, and tenderness of the lower extremities. These are common manifestations of both cellulitis and stasis dermatitis that can mimic the cutaneous changes seen in secondary HOA. However, treatment with antibiotics, topical corticosteroids and emollients, and compression stockings failed to provide symptom relief. The pertinent physical examination findings of digital clubbing in conjunction with joint pain, weight loss, and plantar pain with walking in a patient with recently diagnosed NSCLC made the diagnosis of secondary HOA clear.

Treatment and management. Several treatment modalities have been used for HOA with varying degrees of success (Tables 1 and 2). In general, different therapies can be loosely categorized as either treatment directed toward the underlying cause, such as surgical resection of a primary tumor, and symptomatic management or cosmetic improvement when primary treatment is not possible.[13,49–52]

The initial therapeutic strategy for any patient presenting with secondary HOA should be aimed at treating the inciting cause. There are several documented cases of secondary HOA that have resolved following treatment of an underlying disease.[50–52] While many of these cases relate to pulmonary processes and malignancies, resolution of secondary HOA has been documented following treatment of cyanotic heart disease with surgical correction, liver conditions with grafting and/or transplantation, and esophageal leiomyoma or squamous cell cancer with total esophagectomy.[53–56] These cases demonstrate that while several pathological conditions have been linked to secondary HOA, when the primary cause can be treated, HOA symptoms will most likely improve or resolve.

Unfortunately for many patients including the one presented herein, the underlying problem causing secondary HOA is not amenable to therapy. In cases like this, symptomatic treatment plays an important role in disease management. Nonsteroidal anti-inflammatory drugs (NSAIDs) and octreotide as well as bisphosphonates, such as pamidronate and zoledronic acid have all been used with varying success rates.

Certain NSAIDs, such as rofecoxib, ketorolac, and indomethacin, have been shown to be effective in reducing symptoms related to HOA.[57–59] NSAIDs reduce inflammation by inhibiting cyclooxygenase, an enzyme necessary for the biosynthesis of prostaglandins, thromboxanes, and leukotrienes. While its exact role is unknown, elevated concentrations of prostaglandin E2 (PGE_2) have been demonstrated in patients presenting with symptoms of

TABLE 1. Treatments for primary hypertrophic osteoarthropathy*

TREATMENT	SYMPTOM/S
NSAIDs	Polyarthritis and bone pain
Corticosteroids	Polyarthritis and bone pain
Neurotoxin injections	Facial pachydermia, hyperhidrosis
Beta-blockers	Hyperhidrosis
Glycopyrrolate	Hyperhidrosis
Infliximab**	Polyarthritis and bone pain
CONSIDER REFERRALS TO:	
SPECIALTY	**TREATMENT**
Orthopedics	Surgical correction of clubbing or bony growth
Rheumatology	Treatment of articular pain
Plastic surgery	Excising redundant skin, correcting cutis verticis gyrata
Physical therapy	Improvement in joint function and pain
Opthomology	Surgical or medical correction of eyelid ptosis
Psychiatry	If associated mood disorder given cosmetic appearance and joint disability
Pain management	Treatment of pain

*Thyrotropin and growth hormone levels should be examined to exclude thyroid acropachy and acromegaly
**da Costa FV, de Magalhães Souza Fialho SC, Zimmermann AF, et al.Infliximab treatment in pachydermoperiostosis: a rare disease without an effective therapeutic option.*J Clin Rheumatol.* 2010;16(4):183–184.
Abbreviations: NSAIDs = nonsteroidal anti-inflammatory drugs

TABLE 2. Treatments for secondary hypertrophic osteoarthropathy*

PRIMARY THERAPY
Tumor resection
Radiotherapy
Chemotherapy
Organ transplantation
Treatment of infection or replacement of infected graft
Surgical correction of cyanotic heart disease
SYMPTOMATIC THERAPY
Vagotomay
Adrenergic antagonist
Bisphosphonates (pamidronate or zoledronate)
NSAIDs (rofecoxib, ketorolac, indomethacin)
Octreotide
Gefetinib (EGFR inhibitor)

* May be tried for primary genetic forms, but unlikely successful given different pathogenesis
Abbreviations: NSAIDs = nonsteroidal anti-inflammatory drugs; EGFR = epidermal growth factor receptor

periostitis from HOA.[57] NSAIDs are known to reduce PGE synthesis and it is hypothesized that they exert their clinical efficacy through this mechanism.[57] Kozak et al[57] describe a case of HOA in a 65-year-old woman with recurrent NSCLC and adrenal metastasis who showed clinical improvement with rofecoxib when pain symptoms had been previously refractory to high-dose narcotics. Levels of urinary PGE in the patient correlated with her pain level supporting their theory that COX-derived PGE is involved in the pathogenesis of HOA.

Promising symptomatic treatment of HOA has also been reported with the somatostatin analog, octreotide. Octreotide is a potent inhibitor of growth hormone, glucagon, and insulin that has a well-established role in the treatment of acromegaly and neuroendocrine tumors. Its use in the symptomatic management of secondary HOA has largely been borne out of studies reporting elevated levels of serum growth hormone in patients with HOA from underlying pulmonary malignancy.[2,60,61] Like patients with acromegaly, octreotide's treatment success for HOA may be due to resolving growth hormone dysregulation.[2] A 200µg dose of subcutaneous octreotide daily resulted in complete pain relief in a patient with HOA secondary to squamous cell lung carcinoma.[60] More recently, Angel-Moreno Maroto et al[62]

demonstrated the efficacy of octreotide in achieving complete symptom relief in a young patient following surgical correction of tetralogy of Fallot with pulmonary artery atresia. While its exact mechanism of action is unclear, it has been suggested that the efficacy of octreotide may not only be related to the control of growth hormone, but also to the suggested inhibition of VEGF and endothelial proliferation.[2,46,62]

Newer proposed symptomatic treatments of HOA include the use of bisphosphonates. Traditionally used to treat a variety of bone disorders, such as osteoporosis, Paget's disease, and metastatic bone malignancy, bisphosphonates exert their therapeutic effect through potent inhibition of osteoclastic bone resorption. It remains unclear how this inhibitory effect provides symptomatic relief in patients with secondary HOA. However, many case reports have demonstrated support for the clinical utility of intravenous pamidronate in reducing or resolving symptoms from HOA as well as decreasing scintigraphic evidence of periostitis on bone scan.[5,24,63–67] Speden et al[67] reported three cases of HOA secondary to bronchogenic carcinoma that responded to the use of pamidronate in which pain reduction was reported in all three patients and reduced radiolabel uptake on bone was found in two patients. In two separate case reports, complete resolution of HOA symptoms were reported within 72 hours to 7 days following a 30mg intravenous infusion of pamidronate in one patient with metastatic breast carcinoma and another with cystic fibrosis.[63,64] More recently, zoledronic acid, a bisphosphonate with 40 to 850 fold greater potency than pamidronate, was proposed as treatment with success.[2,24] It has been suggested that not only do bisphosphonates exert their effect on bone metabolism, but also have effects on circulating VEGF. Decreased levels of plasma VEGF has been shown in patients with cancer treated with either pamidronate or zoledronic acids.[2,24]

Often used as therapy for patients with NSCLC, the EDFR tyrosine kinase inhibitor, gefetinib has recently been reported to induce remission of periostitis in a case of HOA with advanced lung adenocarcinoma.[68,69] Given the supporting evidence for the importance of VEGF in the pathogenesis of HOA and its correlation with disease activity, it will be interesting to see if targeted antibodies, such as bevacizumab, used alongside first-line chemotherapies for NSCLC can alleviate pain symptoms in patients with secondary HOA.[70]

Aside from primary treatment options and symptomatic therapies, the skin manifestations of HOA may be aesthetically displeasing and disfiguring for some patients. This is particularly an issue for patients with primary HOA in which skin manifestations tend to be more prominent than rheumatological complaints. Given the genetic cause of primary HOA, treatments can be difficult. Plastic surgery may be necessary to remove excessive facial or scalp skin and beta blockers or neurotoxins may be beneficial for associated hyperhidrosis. Recently, botulinum toxin type A has been used for the treatment of leonine facies (pachydermia) in three patients with pachydermoperiostosis (primary HOA) with modest improvement in cosmetic facial appearance.[71]

CONCLUSION

Although rare, HOA is a relatively straightforward clinical diagnosis for the dermatologist to make, given the clinical characteristics of increased bone tissue at the distal extremities, digital clubbing, and pain associated with underlying disease, particularly lung cancer. In many cases, treatment of the underlying malignancy or chronic medical condition can be curative. However, in the setting of refractory disease, alternative symptomatic treatments must be considered to address the disabling pain reported by patients. Options include opiates, NSAIDs, octreotide, bisphosphonates, and growth factor inhibitors. Although significant strides have been made in the therapeutic options available for the management of HOA over the last several decades, developing a standard protocol will require in-depth basic science research into the disease pathophysiology and clinical trials to compare treatments. Nonetheless, this report emphasizes the variable presentation of this syndrome and demonstrates the benefit of bisphosphonates for symptomatic relief in secondary HOA from pulmonary malignancy.

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